Dynamic Modeling and Management of Tail Risk:

Static, Rolling, and Autoregressive EVT Approaches Applied to Financial Returns

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Abstract

Understanding and accurately quantifying tail risk is critical for financial risk management, particularly during periods of market stress. Traditional Value-at-Risk (VaR) models based on Normal or Student-t assumptions often fail to capture the dynamic nature of financial returns, leading to potential underestimation of extreme losses. This paper explores a range of models for tail risk estimation, including static parametrics VaR, static Generalized Pareto Distribution (GPD), Monte Carlo simulation using GPD, rolling GPD model, and the recently proposed Autoregressive Extreme Value (AEV) model. Using XLF ETF data from 2010 to 2024, the models are calibrated and evaluated based on backtesting metrics, particularly the Kupiec Proportion of Failures (POF) test. The results show that static models systematically misrepresent risk, while rolling GPD models improve calibration at the cost of conservatism. AEV models offer promising adaptability but require further tuning for robust performance across different regimes. These findings align with the broader evidence from financial behavior literature emphasizing that risk tolerance and exposure evolve dynamically over time (Meziani & Noma, 2018). The paper concludes by recommending a dynamic modeling approach for managing financial risk in volatile environments.

Keywords: tail risk, value-at-risk, expected shortfall, extreme value theory (EVT), rolling windows models, financial risk management

Introduction

Accurately modeling the risk of extreme financial losses remains a central challenge in risk management, especially during periods of market stress. Standard risk models, such as Value-at-Risk (VaR), have been criticized for their reliance on restrictive assumptions, including normally distributed and independent returns (Jorion, 2009). Such assumptions routinely break down during crises, leading to substantial underestimation of risk and exposing institutions to catastrophic losses.

The failure of traditional risk measures during systemic events such as the 2008 Global Financial Crisis and the COVID-19 pandemic has motivated a search for more robust frameworks. Extreme Value Theory (EVT) provides a natural alternative by focusing specifically on the tails of the return distribution, offering tools to quantify rare but impactful events (McNeil, Frey, & Embrechts, 2005). However, standard EVT models often assume static tail behavior, which may be insufficient in non-stationary financial environments where risk dynamics evolve over time. Recent developments propose dynamic models that better account for temporal changes in tail behavior. Among them, the Autoregressive Extreme Value (AEV) model introduces a time-varying structure for tail risk, capturing the persistence and clustering of extreme losses (Feng, 2020). Meanwhile, behavioral finance research also suggests that risk tolerance itself evolves with market conditions (Meziani & Noma, 2018), reinforcing the case for adaptive approaches to risk modeling.

This paper investigates multiple approaches to modeling tail risk under both static and dynamic frameworks. Using daily returns and forward-looking drawdowns for the XLF ETF from 2010 to 2024, we systematically implement and evaluate traditional VaR models, static GPD fitting, Monte Carlo tail simulations, rolling GPD models, and the dynamic AEV model. The models are assessed based on their ability to accurately capture tail risk and adapt to changing market conditions, providing insights into the strengths and limitations of each approach. Building on these theoretical insights, the following section outlines the methodological framework employed to empirically evaluate static, rolling, and dynamic EVT models using financial data.

Literature Review

Financial risk management has historically relied on Value-at-Risk (VaR) models to quantify potential portfolio losses. While VaR remains a standard industry metric, traditional models often assume that financial returns are normally distributed and independent over time, assumptions which break down during market crises (Jorion, 2009). Danielsson and de Vries (2000) showed that such models systematically underestimate the probability and magnitude of extreme losses, particularly when returns exhibit heavy tails and volatility clustering.

In response to the limitations of Gaussian assumptions, Extreme Value Theory (EVT) emerged as a leading alternative framework. EVT focuses specifically on modeling the tail behavior of distributions, offering tools to better quantify the risk of rare but severe events (Embrechts, Klüppelberg, & Mikosch, 1997; McNeil, Frey, & Embrechts, 2005). The Generalized Pareto Distribution (GPD), widely used within EVT, characterizes the distribution of exceedances over a high threshold and provides more accurate tail estimates than standard parametric models (Gilli & Këllezi, 2006).

However, most applications of EVT have historically assumed static tail parameters, implying that the severity and frequency of extreme events remain constant over time. This is a problematic assumption, particularly as empirical evidence suggests that financial markets undergo regime shifts where tail risk varies dynamically (McNeil et al., 2005).

To partially address non-stationarity, rolling window approaches have been employed. By recalibrating GPD parameters over moving samples, rolling EVT models attempt to reflect recent market conditions (Feng, 2020). Nevertheless, these models still lack explicit modeling of temporal dependence across exceedances: each window is treated independently, potentially missing persistent shifts in tail behavior.

Recent advances propose more sophisticated dynamic structures. In particular, Feng (2020) introduced the Autoregressive Extreme Value (AEV) model, wherein the GPD scale parameter evolves over time according to an autoregressive process based on previous exceedances. This allows the model to capture clustering and persistence in extreme losses — phenomena often observed during financial crises but inadequately modeled by static or rolling frameworks.

Complementing these statistical innovations, behavioral finance research highlights that risk preferences themselves evolve dynamically. Meziani and Noma (2018) provide evidence that investors' risk aversion fluctuates with economic and market conditions. Such findings support the need for dynamic models that account not only for volatility clustering but also for changing market sentiment and systemic risk perception.

Despite these developments, empirical studies comparing static EVT, rolling EVT, and autoregressive EVT models remain limited, particularly in simultaneous applications to both returns and drawdowns. Drawdowns, defined as the maximum peak-to-trough decline over a given horizon, offer an alternative perspective on financial losses by focusing on cumulative rather than instantaneous risk. Modeling both returns and drawdowns thus provides a more comprehensive view of tail risk exposure. This paper extends the literature by systematically implementing and evaluating static GPD, rolling GPD, Monte Carlo GPD simulation, and AEV models across both return-based and drawdown-based risk measures, using financial data from the XLF ETF between 2010 and 2024.

Data Description

This study utilizes daily adjusted closing price data for the Financial Select Sector SPDR Fund (XLF), a widely followed exchange-traded fund (ETF) that tracks the financial sector of the S&P 500. The dataset spans the period from January 1, 2010 to March 31, 2024, providing approximately 3,560 observations. Data was retrieved from Yahoo Finance using the *yfinance* Python library.

The XLF ETF was selected due to its sector-specific exposure to banks, insurance companies, and diversified financials, which are historically sensitive to systemic market stress and therefore suitable for tail risk analysis. The time frame includes multiple volatility regimes—such as the European sovereign debt crisis, the COVID-19 crash, and recent post-pandemic rate tightening cycles—providing a diverse environment to evaluate risk model performance under stress.

Daily log returns were computed from adjusted closing prices and forward-looking 30-day drawdowns were calculated to serve as alternative risk measures. Observations with missing values were excluded, and no further preprocessing was applied.

Method

1. Defining Financial Risk Measures

1.1 Returns

Daily returns are defined as the logarithmic change in the closing price of the asset:

$$r_t = log\left(\frac{P_t}{P_{t-1}}\right)$$

where P_t is the closing on day *t*. Returns offer a normalized and stationary framework for modeling asset price movements and are commonly used in risk management for calculating instantaneous losses.

1.2 Drawdowns

The 30-day forward maximum drawdown at time t is defined as the largest relative peak-to-trough decline within the next 30 trading days:

$$Drawdown_t = \frac{max(P_{t+1}, ..., P_{t+30}) - P_t}{P_t}$$

Unlike returns, drawdowns capture cumulative and compounding losses over a period, providing an alternative measure of downside risk from an investor's perspective.

2. Static Risk Models

2.1 Historical Empirical VaR

The empirical VaR at a confidence level p is calculated from historical returns by identifying the quantile:

$$VaR_{quantile}(p) = Quantile_p(r_t)$$

where $Quantile_p$ denotes the $1 - p$ quantile.

2.2 Parametric VaR: Normal and Student-t Distributions

Under parametric assumptions, VaR can be expressed as:

Normal VaR:

$$VaR_{Normal}(p) = \mu + \sigma\phi^{-1}(p)$$

where μ and σ are the sample mean and standard deviation of returns, and ϕ^{-1} is the inverse standard normal cumulative distribution function.

Student-t VaR:

Assuming returns follow a Student-t distribution with ν degrees of freedom:

$$VaR_{Studen-t}(p) = \mu + \sigma t_{\nu}^{-1}(p)$$

where t_{ν}^{-1} is the inverse cumulative distribution function of the Student-t distribution.

3. Static Extreme Value Theory (GPD)

3.1 Peak Over Threshold Framework

Extreme Value Theory (EVT) models the distribution of excesses over a high threshold u. For observations ***X*** such that ***X*** > u, the conditional distribution is approximated by the Generalized Pareto Distribution (GPD):

$$P(X > u + x \,|\, X > u) = (1 + \xi 5 \frac{x}{\beta})^{-1/\beta}$$

where ξ is the shape parameter controlling tail heaviness, β is the scale parameter.

3.2 Static GPD and ES

Based on fitted GPD parameters, the VaR and Expected Shortfall at level p is estimated as:

$$VaR_{GPD}(p) = u + \frac{\beta}{\xi}((1 - p)^{-\xi} - 1), ES_{GPD}(p) = \frac{VaR_{GPD(p)}}{1-\xi} + \frac{\beta - \xi u}{1 - \xi}$$

Parameter estimation is performed via Maximum Likelihood Estimation (MLE) on the exceedances above threshold u.

4. Monte Carlo Simulation Based on GPD

Using the fitted static GPD parameters, we simulate 100,000 synthetic exceedances to generate a full empirical distribution on extreme losses. From the simulated losses, Monte Carlo VaR and ES estimates are derived empirically. This approach enables stress testing and worst case scenario modeling beyond observed historical data.

5. Rolling GPD Models

To capture changes in tail behavior over time, we apply a rolling window approach. At each point in time:

- A fixed-size historical window (e.g. 500 days) is used.
- GPD parameters (ξ, β) are estimated from exceedances within the window.
- VaR and ES estimates are updated dynamically based on rolling estimates.

This method reflects recent market volatility without assuming full stationarity. Thresholds and window lengths are tuned based on minimizing Kupiec POF test failures and optimizing empirical fit.

6. Autoregressive Extreme Value (AEV) model

The AEV model, introduced by Feng(2020), introduces dynamic dependence in the GPD scale parameter. The recursion is defined as:

$$log(\beta_t) = \varpi + \alpha y_{t-1} + \gamma log(\beta_{t-1})$$

where:

- β_t is the scale parameter at time t
- y_{t-1} is the magnitude of the previous exceedance (or zero if no exceedance occurred)
- ϖ, α, and γ are parameters to be estimated

Here, the model captures memory in extreme events: larger recent exceedances increase the conditional scale and hence future risk.

Parameter estimation is performed via Maximum Likelihood Estimation (MLE), minimizing the negative log-likelihood over the exceedances. The dynamic VaR estimate under AEV is then:

$$VaR_{AEV}(p) = u + \frac{\beta_t}{\xi}((1 - p)^{-\xi} - 1)$$

7. Backtesting and Model Evaluation

Model performance is evaluated using:

- Kupiec Proportion of Failures (POF) test (Kupiec, 1995) to assess statistical calibration
- Violation rates: comparing observed to expected exceedances
- Visual comparison of predicted VaR/ ES against realized returns and drawdowns

Through this framework, we aim to determine which modeling approach provides the most accurate and adaptive estimate of financial tail risk.

Result

We collected daily close prices for the XLF ETF from January 2010 through April 2024, resulting in 3,089 observations. From these prices, two types of financial risk measures were computed: daily returns and 30-day forward maximum drawdowns. Returns capture day-to-day fluctuations, while drawdowns measure the worst cumulative losses over a short future horizon. Both perspectives complementary views on extreme risk behavior. We evaluate model performance based on VaR at the 99% confidence level, Expected Shortfall (ES), observed versus expected violations, and Kupiec POF test statistics.

Table 1: Summary of Tail Risk Estimates and Backtesting Metrics Across Models

Model	Data Type	VaR (99%)	ES (99%)	Expected Violations	Actual Violations	Observed Violation Rate	LR Statistics
Empirical	Returns	0.0375	0.0559	-	-	-	-
Normal	Returns	0.0323	0.0371	-	-	-	-
Student-t	Returns	0.0384	0.0590	-	-	-	-
Static GPD + MC	Returns	-0.1303	-0.1950	-	-	-	-
Rolling	Returns	Rolling	Rolling	30.89	18	0.0058	6.3921

GPD							
Rolling GPD	Drawdowns	Rolling	Rolling	30.60	16	0.0052	-22.8742
AEV	Returns	Rolling	Rolling	30.89	1460	0.4726	-22.9120
AEV	Drawdowns	Rolling	Rolling	30.60	119	0.0389	-22.8742

Across static models, the Normal distribution exhibited the most substantial underestimation of extreme losses, with VaR and ES values notably lower than empirical results. The student-t distribution improved tail fitting modestly but remained inadequate for capturing extreme tail risk.

The static GPD model, incorporated into Monte Carlo simulation, produced highly conservative risk estimates, forecasting substantially larger extreme losses compared to static VaR methods. While this conservatism strengthens risk resilience, it lacks responsiveness to evolving market conditions.

Rolling GPD models demonstrated improved adaptability by recalibrating tail estimates dynamically. Both GPD on returns and on drawdowns exhibited violation rates below the expected 1%, indicating conservative risk estimates. The observed violation rate for rolling GPD on returns was 0.58% and 0.52% for drawdowns, suggesting overcapitalization relative to the theoretical target.

The AEV models, incorporating autoregressive dynamics in the scale parameter, exhibited divergent behavior. On returns, the AEV model significantly underestimated tail risk, with an observed violation rate of 47%. On drawdowns, the model performed somewhat better, though still exceeding the expected violation rate at 3.89%. These results indicate that while dynamic scaling captures some persistence of extremes, model miscalibration during transitions between calm and stressed periods can degrade predictive performance.

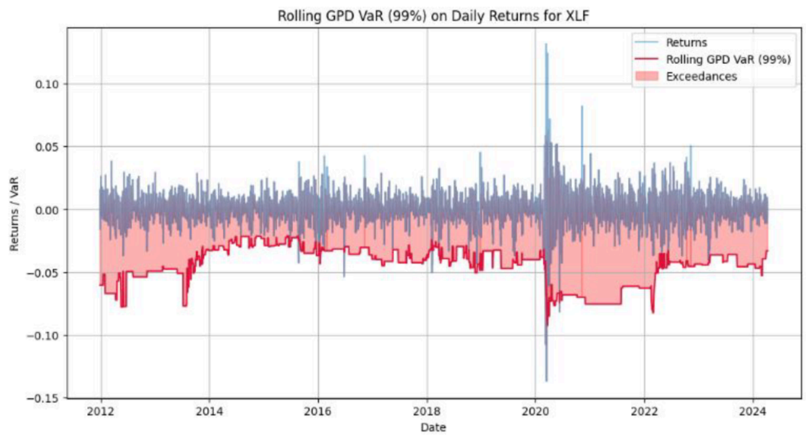

Figure 1: Rolling GPD VaR 99% Forecast and Realized Daily Returns for XLF, 2010-2024

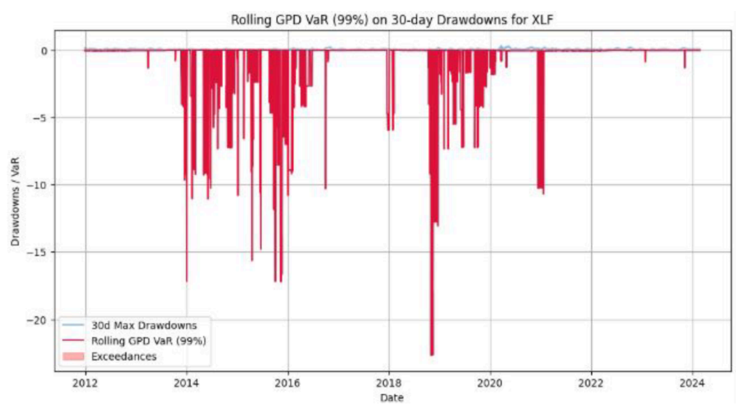

Figure 2: Rolling GPD VaR 99% Forecast and 30-day Forwards Drawdowns for XLF, 2010-2024

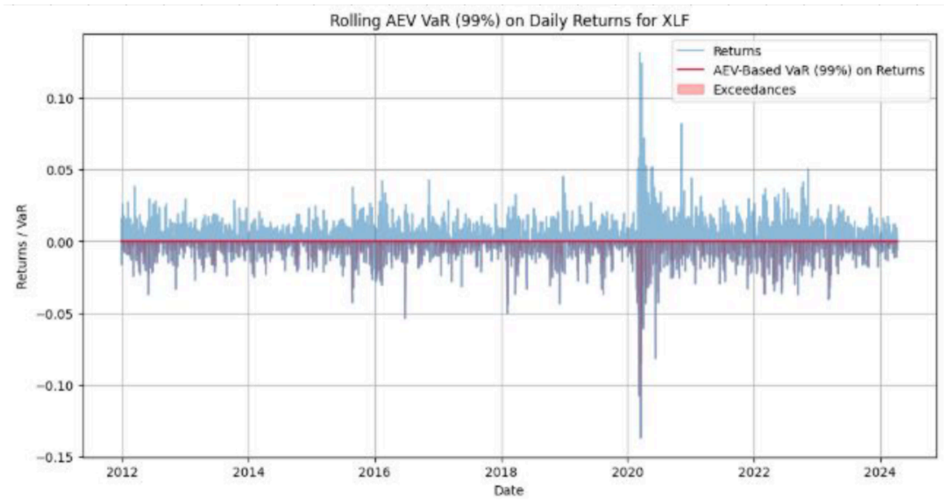

Figure 3: VaR 99% Forecast vs. Realized Daily Returns for XLF

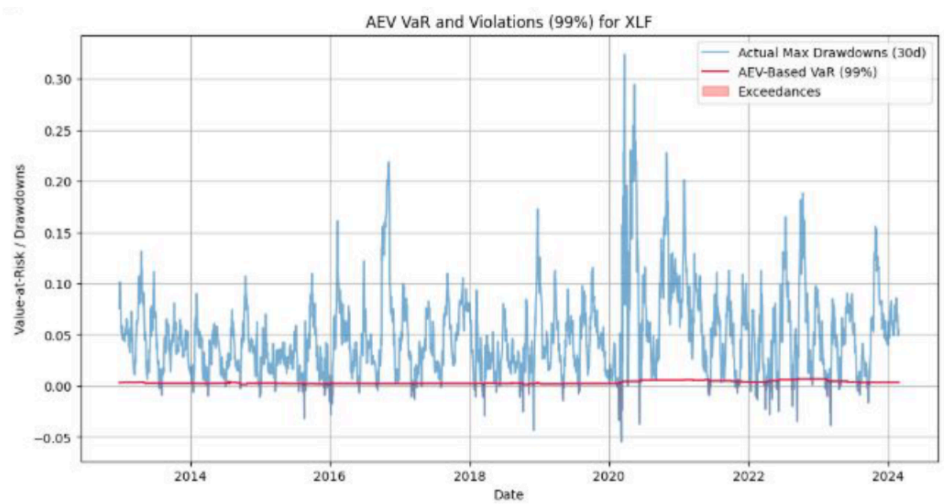

Figure 4: Evolution of the AEV Model's Scale Parameter (β_t) Over Time,

Showing Response to Market Shocks

Discussion

The results of this study reveal substantial differences in the ability of static, rolling, and autoregressive models to capture tail risk dynamics in financial markets.

Static models, based on Normal or Student-t assumptions, consistently underestimated extreme losses compared to observed data. This supports prior findings that Gaussian-based VaR models are prone to underestimation during crisis periods due to their failure to capture fat tails and volatility clustering (Jorion, 2009; McNeil, Frey, & Embrechts, 2005; Danielsson & de Vries, 2000). While Student-t models marginally improved tail estimation due to their heavier tails, they still lacked responsiveness to changing volatility, as discussed in Gençay & Selçuk (2004) and Allen et al. (2012), where static models failed to adapt in high-risk periods.

Extreme Value Theory (EVT), applied via Generalized Pareto Distribution (GPD) fitting and Monte Carlo simulations, produced more conservative estimates. These models effectively emphasized tail heaviness, aligning with Embrechts et al. (1997) and Gilli & Këllezi (2006), but their static nature limited real-time adaptability. Static EVT approaches may thus be better suited for stress testing and regulatory use (e.g., Basel III capital frameworks; see McNeil et al., 2015), rather than short-horizon operational risk control.

Rolling GPD models demonstrated meaningful adaptability by updating tail estimates through recalibrated windows. The models adjusted upward during volatility spikes (e.g., COVID-19), supporting their usefulness in detecting regime changes. However, their consistently low violation rates reflect excessive conservatism, a drawback also noted in studies by Chávez-Demoulin et al. (2016) and Patton et al. (2019). Overestimating risk in tranquil markets can lead to overcapitalization, which is costly for portfolio optimization.

Autoregressive Extreme Value (AEV) models offered a promising direction by capturing temporal dependence in extreme loss behavior. In our application, however, AEV applied to returns produced violation rates far exceeding theoretical expectations—suggesting underestimation of risk. Similar lags in regime detection were observed in simulations by Feng (2020) and recent extensions in Tang et al. (2023). The autoregressive update may lag rapid market shifts, particularly when trained on stable periods. Nonetheless, the model succeeded in reproducing clustering of extremes, showing potential when properly tuned.

Taken together, our findings suggest that no single model suffices for all market conditions. Static parametric models are computationally efficient but lack realism. EVT-based models improve tail sensitivity, yet struggle to respond to market regimes. Rolling GPD estimates strike a practical balance but lean conservative. AEV introduces more realistic dynamics but remains sensitive to tuning and threshold selection. Hybrid approaches—such as combining rolling GPD with autoregressive structures or implementing adaptive thresholds—present an exciting avenue for future research (see Dutta & Perry, 2007; McNeil et al., 2015; Berkes et al., 2022).

Conclusion

This paper examined the performance of static, rolling, and dynamic models for estimating and managing tail risk in financial markets. Using daily returns and 30-day forward drawdowns from the XLF ETF, we applied traditional VaR methods, static EVT-based models, Monte Carlo simulations, rolling GPD models, and the Autoregressive Extreme Value (AEV) framework.

The findings highlight that static models consistently underestimate extreme risks, failing to account for the heavy tails and dynamic shifts observed in financial returns. Static EVT models, while conservative, lack adaptability to changing market regimes. Rolling GPD models offered improved responsiveness but tended toward conservatism, potentially resulting in excessive capital allocation during stable periods. AEV models introduced valuable dynamic scaling but faced challenges in calibration, leading to risk underestimation during high-volatility episodes.

Overall, the study underscores the importance of dynamic approaches in modeling financial tail risk. While no single model provided a perfect solution, combining the adaptability of rolling methods with the dynamic scaling properties of autoregressive frameworks holds promise for more robust risk management strategies. Future work may further explore hybrid models, regime-switching approaches, or machine learning-based dynamic threshold selection to enhance the predictive performance of tail risk estimators in volatile markets.

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Appendix

All code used in this paper is available at: https://github.com/valle1306/Dynamic-Tail-Risk

The repository contains two Jupyter notebooks:

- EDA_&_AEV.ipynb: Implements the AEV model on 30-day forward drawdowns.
- GPD_VaR_&_ES_Implementation.ipynb: Implements static and rolling GPD models for tail risk estimation.